

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 1**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
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Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A particle undergoes simple harmonic motion described by $x(t) = 0.05 \cos(6\pi t + \pi/4)$, where x is in metres and t in seconds. What is the period of oscillation? [2 marks]

- A. $\frac{1}{6}$ s
- B. $\frac{1}{3}$ s
- C. 3 s
- D. 6π s

2. A block of mass 0.4 kg is attached to a spring with spring constant $k = 160$ N/m and oscillates with amplitude $A = 0.08$ m. What is the maximum speed of the block? [2 marks]

- A. 0.80 m/s
- B. 1.60 m/s
- C. 3.20 m/s
- D. 6.40 m/s

3. A damped oscillator has mass $m = 0.25$ kg, spring constant $k = 100$ N/m, and damping coefficient $b = 10$ kg/s. The damping parameter is $\gamma = b/m$ and the natural frequency is $\omega_0 = \sqrt{k/m}$. What is the damping regime of this system? [2 marks]

- A. Underdamped ($\gamma < 2\omega_0$)
- B. Critically damped ($\gamma = 2\omega_0$)
- C. Overdamped ($\gamma > 2\omega_0$)
- D. Undamped ($\gamma = 0$)

4. A lightly damped oscillator is driven by an external sinusoidal force. At resonance, which of the following statements is correct? [2 marks]

- A. The amplitude is at its maximum and the phase lag of the displacement behind the driving force is $\pi/2$.
- B. The amplitude is at its maximum and the displacement is in phase with the driving force.
- C. The amplitude is zero and the phase lag is π .
- D. The amplitude is at its maximum and the phase lag is π .

5. A guitar string has a linear mass density of $\mu = 0.004 \text{ kg/m}$ and is held under a tension of $T = 100 \text{ N}$. What is the speed of transverse waves on this string? [2 marks]

- A. 25 m/s
- B. 50 m/s
- C. 158 m/s
- D. 250 m/s

6. Two strings are joined together. The impedance of string 1 is Z_1 and the impedance of string 2 is Z_2 . Under what condition is there **no** reflected wave when a wave travelling on string 1 reaches the junction? [2 marks]

- A. $Z_1 = 0$
- B. $Z_2 = 0$
- C. $Z_1 = Z_2$
- D. $Z_1 Z_2 = 1$

7. A jet engine at a certain distance produces a sound intensity of $I = 0.01 \text{ W/m}^2$. What is the corresponding sound level in decibels? (Take $I_0 = 10^{-12} \text{ W/m}^2$.) [2 marks]

- A. 80 dB
- B. 90 dB
- C. 100 dB
- D. 120 dB

8. An ambulance siren emits sound at a frequency of $f = 600 \text{ Hz}$. The ambulance is approaching a stationary observer at a speed of $v_s = 34 \text{ m/s}$. Taking the speed of sound in air as $v = 340 \text{ m/s}$, what frequency does the observer hear? [2 marks]

- A. 540 Hz
- B. 600 Hz
- C. 660 Hz
- D. 667 Hz

9. A string of length $L = 0.80 \text{ m}$ is fixed at both ends and supports transverse waves with speed $v = 320 \text{ m/s}$. What is the fundamental frequency of vibration? [2 marks]

- A. 100 Hz
- B. 200 Hz
- C. 400 Hz

D. 640 Hz

10. Two tuning forks are sounded together. Fork A has a frequency of 440 Hz and fork B has a frequency of 446 Hz. How many beats per second are heard? [2 marks]

A. 3

B. 6

C. 443

D. 886

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. Which of the following expressions represents a **linear operator** acting on a Hilbert space? [2 marks]

- A. $\langle\alpha|\beta\rangle$
- B. $|\alpha\rangle\langle\beta|$
- C. $|\alpha\rangle$
- D. $\langle\beta|$

2. A quantum state is written as $|\psi\rangle = \alpha(|1\rangle + 2|2\rangle)$, where $|1\rangle$ and $|2\rangle$ are orthonormal basis states and α is a real, positive normalisation constant. What is the value of α ? [2 marks]

- A. $1/3$
- B. $1/\sqrt{3}$
- C. $1/\sqrt{5}$
- D. $1/5$

3. Let $|\phi\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{i}{\sqrt{2}}|2\rangle$ and $|\psi\rangle = \frac{1}{\sqrt{2}}|1\rangle - \frac{i}{\sqrt{2}}|2\rangle$, where $\{|1\rangle, |2\rangle\}$ is an orthonormal basis. What is $\langle\phi|\psi\rangle$? [2 marks]

- A. 0
- B. 1
- C. i
- D. $-i$

4. Consider the following operators defined on a two-state orthonormal basis $\{|1\rangle, |2\rangle\}$. Which one is Hermitian? [2 marks]

- A. $\hat{A} = i|1\rangle\langle 2| + i|2\rangle\langle 1|$
- B. $\hat{B} = |1\rangle\langle 2| - |2\rangle\langle 1|$
- C. $\hat{C} = |1\rangle\langle 2| + |2\rangle\langle 1|$
- D. $\hat{D} = i|1\rangle\langle 2| - i|2\rangle\langle 1|$

5. A particle is in the state $|\psi\rangle = \frac{3}{5}|E_1\rangle + \frac{4}{5}|E_2\rangle$, where $|E_1\rangle$ and $|E_2\rangle$ are orthonormal energy eigenstates with energies E_1 and E_2 respectively. If the energy is measured, what is the probability of obtaining E_1 ? [2 marks]

- A. $3/5$
- B. $9/25$
- C. $16/25$
- D. $4/5$

6. A spin-1/2 particle is prepared in the state $|\uparrow_z\rangle$. A measurement of S_x is performed. What is the probability of obtaining $+\hbar/2$? [2 marks]

- A. 0
- B. $1/4$
- C. $1/2$
- D. 1

7. A quantum system has Hamiltonian H . A state is prepared at $t = 0$ and then evolves in time under H . Which of the following quantities is **always** conserved during this time evolution? [2 marks]

- A. The probabilities of measuring each position eigenvalue
- B. The expectation value of S_x
- C. The probability of measuring each energy eigenvalue
- D. The expectation value of S_z

8. Consider two spin-1/2 particles, labelled 1 and 2. Which of the following two-particle states is **entangled**? [2 marks]

- A. $|\uparrow_z\rangle_1 |\downarrow_z\rangle_2$
- B. $\frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 + |\downarrow_z\rangle_1) |\uparrow_z\rangle_2$
- C. $\frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 |\downarrow_z\rangle_2 - |\downarrow_z\rangle_1 |\uparrow_z\rangle_2)$
- D. $|\downarrow_z\rangle_1 |\downarrow_z\rangle_2$

9. Two identical fermions occupy a joint quantum state $|\Psi\rangle$. What happens to $|\Psi\rangle$ when the labels of the two particles are exchanged (i.e. the SWAP operator is applied)? [2 marks]

- A. $|\Psi\rangle$ is unchanged: $\text{SWAP} |\Psi\rangle = +|\Psi\rangle$
- B. $|\Psi\rangle$ acquires a factor of -1 : $\text{SWAP} |\Psi\rangle = -|\Psi\rangle$
- C. $|\Psi\rangle$ acquires a factor of i : $\text{SWAP} |\Psi\rangle = i|\Psi\rangle$
- D. $|\Psi\rangle$ becomes orthogonal to the original state

10. A neutron is a composite particle made of quarks. What is the quark content of a neutron? [2 marks]

A. uuu

B. uud

C. udd

D. ddd

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

Projectile Motion and Energy Conservation

A ball is launched from ground level at speed $v_0 = 25$ m/s at an angle $\theta = 53^\circ$ above the horizontal. Take $g = 9.81$ m/s².

- (a) Find the maximum height H reached by the ball. 4
- (b) Find the range R of the ball (the horizontal distance when it returns to ground level).
3
- (c) A second ball of mass $m = 0.2$ kg is dropped from rest at height H (the value found in part (a)). It falls onto a vertical spring of spring constant $k = 500$ N/m resting on the ground. Using conservation of energy, find the maximum compression x of the spring. 4
- (d) The fully compressed spring is now used to launch a marble of mass $m_{\text{marble}} = 0.1$ kg horizontally from the top of a 20 m high cliff. Using the compression found in part (c), find the speed of the marble at the instant it hits the ground. 4

Question 2.

[15 marks]

Thermodynamic Processes

Two moles ($n = 2$) of a monatomic ideal gas are initially at temperature $T_1 = 300$ K and volume $V_1 = 0.05$ m³. Take $R = 8.314$ J/(mol·K), $\gamma = 5/3$ for a monatomic ideal gas, and $C_V = \frac{3}{2}R$ per mole.

- (a) The gas undergoes a **reversible isothermal expansion** to volume $V_2 = 0.15$ m³. Calculate the work done by the gas. 3
- (b) Calculate the heat absorbed by the gas during this isothermal expansion. 3
- (c) The gas then undergoes a **reversible adiabatic expansion** from ($T_2 = 300$ K, $V_2 = 0.15$ m³) to a final volume $V_3 = 0.30$ m³. Find the final temperature T_3 . 4
- (d) Calculate the work done by the gas during the adiabatic expansion. 2
- (e) Calculate the total entropy change of the universe for the isothermal process in part (a). Comment on whether this result is consistent with the process being reversible. 3

Question 3.

[15 marks]

Damped and Forced Oscillations

- (a) Write down the equation of motion for a damped harmonic oscillator of mass m , spring constant k , and damping coefficient b (where the damping force is $-b\dot{x}$). Define the damping parameter γ in terms of b and m . 2
- (b) A block of mass $m = 0.5$ kg is attached to a spring with spring constant $k = 200$ N/m and experiences a damping coefficient $b = 2$ kg/s. Calculate the natural angular frequency ω_0 , the damping parameter γ , and the damped angular frequency ω_d . 3
- (c) Calculate the quality factor Q of this system. State whether the system is underdamped, critically damped, or overdamped, giving a brief justification. 2
- (d) The system is now driven by an external force $F(t) = F_0 \cos(\omega t)$ with $F_0 = 5$ N. Find the steady-state amplitude of oscillation when the driving angular frequency is $\omega = 19$ rad/s. 4
- (e) Find the driving angular frequency ω_{res} at which the steady-state amplitude is maximum (the amplitude resonance frequency). Calculate the maximum amplitude A_{max} at this resonance frequency. 4

Question 4.

[15 marks]

Spin-1/2 and Measurement

A spin-1/2 particle is prepared in the state

$$|\psi\rangle = \frac{1}{\sqrt{3}} |\uparrow_z\rangle + \sqrt{\frac{2}{3}} |\downarrow_z\rangle.$$

- (a) Verify that $|\psi\rangle$ is normalised. If S_z is measured, state the possible outcomes and their probabilities. 3
- (b) Express $|\psi\rangle$ in the S_x eigenbasis $\{|\uparrow_x\rangle, |\downarrow_x\rangle\}$. Hence, if S_x is measured on the original state $|\psi\rangle$, find the probability of each outcome. 4
- (c) After measuring S_x and obtaining the result $+\hbar/2$, the particle's state collapses to $|\uparrow_x\rangle$. If S_z is then measured on this post-measurement state, find the probability of each outcome. 4
- (d) Consider the operator $\hat{A} = \hbar(|\uparrow_z\rangle\langle\downarrow_z| + |\downarrow_z\rangle\langle\uparrow_z|)$. Find the eigenvalues and normalised eigenvectors of $\hat{A}$. Determine whether $\hat{A}$ is Hermitian, justifying your answer. 4

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m